

Same Product, Different Visibility: Representation Robustness in Neural E-Commerce Retrieval

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Abstract

Neural retrieval increasingly mediates which Web content can be discovered, compared, or used by agentic interfaces. Semantically equivalent information should ideally receive comparable retrieval opportunities, yet structured records are routinely flattened into ordered text. We study this representation robustness problem in product search by holding facts and relevance fixed while permuting attributes. Across two benchmarks and multiple retrievers, 4.4–13.3% of highest-relevance pairs cross Top-20 under catalog-wide permutations; target-only interventions confirm that the same product can change visibility with every competitor fixed. We then test whether two public, query-blind visibility-rewriting strategies withstand the same intervention. On frozen samples, rewriting raises rather than removes order sensitivity and reduces BGE target inclusion on ESCI by 4.9–12.7 percentage points. Strict field checks reveal extensive fact loss and output truncation, so these adapted local-model results do not reproduce or refute the original rewriting method. Canonical preprocessing eliminates incoming-order variation, but guarded pipelines yield no gain over canonical text and full-catalog recall shows a significant 1.8-point loss in one of four dataset–encoder cells. Generation replicates further show that stochastic variation can rival the measured order effect. Neural Web retrieval should therefore evaluate visibility optimization jointly for relevance, representation robustness, factuality, and candidate cost.

CCS Concepts

• **Information systems** → **Information retrieval**; *Recommender systems*.

Keywords

Web retrieval, product search, robustness, representation invariance, visibility

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1 Introduction

Web users increasingly reach information through systems that decide what to retrieve before the user sees it. Search, recommendation, retrieval-augmented systems, and emerging agentic interfaces all depend on an initial selection of content. For a shopping agent to compare a relevant product, that product must first enter its candidate set. This motivates a retrieval question, without assuming that better retrieval necessarily improves recommendations or purchases: *can irrelevant choices in how unchanged information is represented determine whether relevant content is found?*

Much of the Web is structured or semi-structured. Products, jobs, hotels, and events are stored as fields, yet neural retrieval often

consumes flattened token sequences. Many underlying facts have no meaningful linear order. Product color, material, dimensions, and compatibility remain the same facts when their positions in a record change. This mismatch makes serialization a potential nuisance signal between semantic relevance and retrieval visibility.

E-commerce provides a controlled setting for studying that mismatch. We preserve product identity, all source facts, the query, the model, and the relevance labels while permuting complete attribute atoms. We distinguish a catalog-wide intervention, corresponding to a shared indexing change, from a target-only intervention that holds every competitor in its original representation. The latter supports the precise comparison: same product, same facts, same competitors, same query, same model; only the target serialization changes.

Figure 1 illustrates the effect. For *turquoise chair*, an Exact-relevant product moves between ranks 5 and 1,527 under different attribute orders with every competitor fixed. Its original rank is 486. The example is selected for interpretability and a large crossing; aggregate results, rather than this extreme case, establish the population pattern. The full title, source atoms, identifiers, and all tested ranks are included in the artifact and appendix.

Query: "turquoise chair"

Same Exact-relevant product and facts
Same model • Every competitor fixed in C0

Order A (C2s4)	Order B (C2s1)
Rank 5	Rank 1,527
VISIBLE at Top-20	HIDDEN at Top-20

Only the target attribute order changes
Should serialization determine visibility?

Figure 1: Target-only intervention, WANDS query 162 and product 34536, MiniLM. The same Exact-relevant product crosses Top-20 with competitors fixed in C0. This differs from catalog-wide reserialization.

Order instability is important, but eliminating it is not sufficient. A relevant product that is never retrieved under any representation has zero crossing instability. The same distinction applies to content optimization. A rewritten listing can rank well under one input order while remaining brittle to another, and a stable rewrite can consistently miss the candidate set. We therefore ask: *do existing visibility-oriented rewriting strategies preserve relevant-product inclusion when the same source facts arrive in different orders, and is canonical preprocessing sufficient?*

The available intervention also matters. A merchant can change one product’s text, whereas a platform can change an entire index, aggregate embeddings, or store multiple vectors per product. A target-only gain does not establish a market-wide benefit, and a platform-only method is not a merchant-deployable rewriting rule. We compare these settings explicitly and prevent multi-view retrieval from receiving extra candidate slots: all methods return the same number of unique products.

This question complements recent generative-engine optimization (GEO) work. E-GEO rewrites one product inside a fixed ten-product candidate set and evaluates an LLM reranker; its authors explicitly separate and cache retrieval. SAGEO Arena already broadens GEO evaluation to retrieval, reranking, and generation. We neither dispute their reranking results nor claim the first end-to-end GEO study. Instead, we isolate a different nuisance variable before reranking: the order of unchanged structured product facts.

Our study contributes four pieces of evidence. First, we retain cross-model, cross-dataset and target-only evidence that raw attribute order changes relevant-product inclusion, including products whose complete text is not truncated. Second, we cross two public E-GEO prompts with seven input orders under a query-blind local rewriter, separating average inclusion from order sensitivity. Third, we audit generated facts, estimate same-input generation variation, and evaluate a prespecified canonical fallback rather than treating every rewrite as meaning-preserving. Fourth, we compare these merchant-side rewrites with simple canonicalization and existing platform-side hybrid, centroid, set, and candidate-budget controls. Negative transfer and factuality results remain part of the claim.

2 Related Work

2.1 Structured Product and Hybrid Retrieval

Product search must match queries to heterogeneous catalog fields. Lexical retrieval such as BM25 provides a strong order-insensitive baseline under a fixed token multiset [16]; dense matching and field-aware models address semantic mismatch [6]. Attribute extraction can supply explicit matching signals [12]. CHARM constructs hierarchical field representations with block-triangular attention and uses aggregated vectors for candidate retrieval followed by richer field-level matching [8]. Its learned, field-aware hierarchy differs from our frozen-encoder views of complete, fact-equivalent product records. We use it to position multi-vector product retrieval, not as an implemented baseline.

Hybrid retrieval combines lexical and semantic signals. Reciprocal rank fusion combines rankings without requiring comparable score scales [7]. We include fixed full-list RRF as a practical competitor because a representation strategy is less compelling if a simple hybrid already obtains better recall. Our comparisons hold candidate count fixed and separately report the extra storage and scoring cost of multiple product vectors. The purpose is to evaluate the representation decision, rather than propose another fusion algorithm.

2.2 Content and Visibility Optimization

GEO studies how content changes influence visibility in generative systems [2]; ranking-incentivized modification similarly studies

strategic changes under quality constraints [9]. E-GEO specializes GEO to e-commerce and releases fifteen hand-crafted prompts plus meta-optimized prompts [3]. Its query-blind rewriter changes one product description, while a fixed set of ten products is reranked by LLMs. Its implementation caches retrieval and therefore holds that candidate set fixed. That controlled setting answers a reranking question; it does not estimate whether a rewrite still reaches the candidate set under different source serializations.

SAGEO Arena addresses the broader pipeline by dynamically retrieving, reranking, and generating over structured Web documents [11]. It reports that body-text optimization can degrade retrieval and that structural fields can mitigate stage-specific failures. Our contribution is therefore narrower than a first end-to-end analysis: we vary only the order of complete product atoms, evaluate relevant-target inclusion with competitors fixed, verify source facts, and separate input-order effects from generation repeats. We use two released E-GEO prompts with a local model and label them adapted baselines; we do not reproduce E-GEO’s GPT-4.1 meta-optimization or LLM-ranker scores.

2.3 Representation Robustness

Serialization sensitivity is not a new discovery in neural retrieval. Bhandari et al. study alternative table serializations, centroid representations, and a residual adapter toward centroid targets [4]. Their centroids motivate our full-record averaging baseline, which is distinct from independent attribute encoding followed by set pooling. We extend the empirical question to judged-relevant product inclusion, target-only causal control, persistent omission, cross-encoder candidate limitations, and recall–cost tradeoffs. We do not claim that a finite mean over views is a new invariant-learning algorithm.

Order effects and behavioral retrieval testing also caution against treating aggregate effectiveness as a complete description of a neural system [1, 13]. Our task-defined equivalence class is deliberately narrower than arbitrary rewriting: complete raw attribute atoms are moved while their contents and multiplicities remain fixed. Generalization to other structured Web domains remains an empirical question.

3 Problem Formulation

Let $F(p)$ denote the information underlying item p , and let $\mathcal{R}(F(p))$ contain representations that preserve that information for the task. Human relevance $\text{Rel}(q, p)$ remains fixed across the class. A normalized dual encoder scores a representation by $z(q, r(p)) = \langle E_q(q), E_p(r(p)) \rangle$. A representation-robust retriever should ideally avoid substantial visibility differences induced solely by task-irrelevant changes in r .

For rank $\rho_r(q, p)$ and cutoff K , define

$$\text{VI@}K(q, p) = \mathbb{I} \left[\min_{r \in \mathcal{R}} \rho_r(q, p) \leq K < \max_{r \in \mathcal{R}} \rho_r(q, p) \right]. \quad (1)$$

VI is an any-in/any-out event over the tested family, not a single-update loss probability. We also partition relevant pairs into always retrieved, sometimes retrieved, and never retrieved. Sometimes retrieved equals crossing; never retrieved can be a serious recall failure despite $\text{VI}=0$. Rank range describes movement independently of a cutoff. VI need not increase with K , since a boundary can move beyond an item’s entire rank interval.

Let H_q be the judged highest-relevance products for query q . Our primary strategy metric is

$$\text{Recall@K}(q) = \frac{|H_q \cap \text{TopK}(q)|}{|H_q|}. \quad (2)$$

Queries with $|H_q| = 0$ have undefined recall and are excluded only from recall aggregation. Query-macro averages weight eligible queries equally; pair-micro statistics weight relevant pairs equally. For each method–Original comparison we retain rescued and newly missed relevant pairs; their difference, normalized by $|H_q|$, is the per-query recall change.

Relevance–visibility alignment requires useful retrieval as well as stability. We report condensed NDCG, which removes unjudged products before discounting, separately from complete-catalog recall and rank visibility [10, 17]. Unknown labels are not treated as irrelevant. In a two-stage pipeline, representation-dependent Irrecoverable@100 counts products present in some first-stage pools but absent in others; persistent omission is reported separately. Reranker VI on common candidates is conditional on survival and cannot be compared directly with all-highest-pair VI.

4 Interventions and Retrieval Strategies

4.1 Equivalent Inputs and Intervention Permissions

The primary family contains C0 source order, C1 reversal, and five deterministic attribute permutations. WANDS atoms are raw key–value fields; ESCI uses available brand, color, and complete bullet-point strings as atoms, retaining the internal bullet text. Audits preserve complete raw atoms and multiplicity, non-attribute fields, source values, and character/lexical token multisets. Catalog-wide intervention changes all products. Target-only intervention changes one relevant target at a time with every competitor in C0:

$$\rho_s^{\text{target}}(q, p) = 1 + \sum_{u \neq p} \mathbb{I}[z(q, C0(u)) > z(q, s(p))], \quad (3)$$

with stable catalog-order tie handling. The old target entry is removed before insertion. These interventions answer different questions; their crossing rates do not form an additive causal decomposition.

Merchant-side strategies must produce one complete product text without knowledge of a future query. We freeze four field-priority rules emphasizing type/style, appearance, use/compatibility, or dimensions/material. They move entire raw atoms, retain the original free text and section order, and use a canonical lexical fallback. Matching uses field keys where available and literal prefixes for unkeyed ESCI atoms. One global rule is chosen on development Recall@100; cNDCG and a fixed simplicity order break ties. No product-specific or category-specific supervised policy is fitted.

We distinguish pure raw-atom sorting from the historical canonical template M2, which also adds field framing and changes section placement. Both are retained as baselines. This avoids attributing the template’s effect entirely to sorting. The selected priority rule is evaluated both catalog-wide and target-only; the latter is a diagnostic of a single record’s effect, not a deployment policy that knows test relevance labels.

4.2 Optimization under Input-Order Intervention

Let G be a frozen, query-blind product rewriter and let C sort complete raw attribute atoms lexically while leaving every non-attribute field unchanged. For each source order s , we evaluate

$$E(x_s(p)), \quad E(C(x_s(p))), \quad E(G(x_s(p))), \quad E(G(C(x_s(p))))). \quad (4)$$

The intervention precedes optimization. We do not shuffle sentences produced by G , because a free-text sentence permutation is not guaranteed to preserve meaning. C differs from historical M2: it adds no labels, framing, or section movement. Since $C(x_s(p))$ is identical for all seven orders, canonical-first generation can be cached once. Its resulting zero order-VI is a structural property of canonicalization and caching, not evidence that G itself is invariant.

We study two public E-GEO prompts fixed before test generation: its hand-crafted authoritative heuristic (G_1) and its released meta-optimized technical prompt (G_2). Both explicitly request fact preservation and receive the complete product text but no query, label, score, competitor, or rank. The prompts originally ran with GPT-4.1 against fixed LLM-reranker candidates. We instead use one pinned local instruction model, so these are adapted E-GEO baselines rather than reproduced E-GEO results.

Each highest-label target is replaced independently in the C0 catalog. We remove its old C0 score before inserting the counterfactual score; no favorable target versions coexist. Averaging these binary outcomes gives a *target-intervention inclusion rate*, not Recall from one deployable catalog. We report query-macro and pair-micro forms, seven-order mean inclusion, VI, always/sometimes/never inclusion, and finite-family worst-observed coverage. The minimum of seven global schedule averages and the per-product worst-observed inclusion are distinct estimands.

4.3 Generated-Fact and Randomness Controls

Raw permutations satisfy exact atom and token-multiset equality. Free-text rewrites do not. A prespecified rule checker compares each output with the full source inventory: numbers and units, colors, alphanumeric model strings, negated and compatibility attributes, title coverage, and structured values. Duplicate keys with conflicting values remain in the inventory. We report operational outputs with their failure rate, then a common-valid subset when support exists. A guarded pipeline replaces every failed, empty, or truncated output with raw canonical text; it never selects an output by rank. This automated check is conservative and incomplete, and is neither human review nor semantic ground truth.

Primary generation is greedy with content-addressed caching. On eight hash-selected products, a separate 7 orders $\times$ 3 stochastic-replicate design estimates generation variation. We compute order VI within each replicate and same-input variation across replicates separately; pooling all 21 outputs would incorrectly attribute sampling noise to order.

4.4 Platform-Side Aggregation and Fusion

For full-record multi-view methods, generate nested sets of $m = 2, 4, 7$ views by shuffling a canonical raw-atom list using fixed product-ID-dependent seeds. Each complete record is encoded and

normalized independently. Centroid retrieval uses

$$v_m(p) = \text{norm} \left(\frac{1}{m} \sum_{j=1}^m E_p(r_j(p)) \right), \quad (5)$$

whereas multi-vector retrieval scores p by $\max_j \langle E_q(q), E_p(r_j(p)) \rangle$. Reduction is by product before selecting K unique products. Equal m is used for every product; actual distinct text and encoded-view counts are recorded. The canonical-anchored generator is independent of incoming attribute order. This does not make finite averaging an exhaustive orbit average or establish robustness to other semantic transformations.

The existing set-mean method instead encodes individual attributes. With normalized non-attribute vector $t(p)$, it computes

$$a(p) = \text{norm} \left(\frac{1}{|A(p)|} \sum_{a \in A(p)} E_p(a) \right), \quad (6)$$

$$v(p) = \text{norm}(.5t(p) + .5a(p)).$$

The attribute mean is normalized before interpolation, matching the implementation; the earlier draft omitted this intermediate step. Empty attribute sets contribute zero. This proof of concept removes order dependence but may lose useful cross-attribute interactions. It is evaluated for recall and effectiveness, not declared successful because VI is zero.

BM25 uses the full lowercase alphanumeric token multiset without positional features or truncation. Hybrid retrieval adds full-list reciprocal-rank scores with constant 60 [7]. These are platform controls. Multi-vector and centroid methods increase offline encoding; only the former retains multiple index vectors. None is represented as a merchant-editable text strategy.

5 Experimental Setup and Audit

5.1 Data, Splits, and Evaluation History

The historical WANDS catalog contains 42,994 products, 480 queries, and 231,859 deduplicated, noncontradictory judgments. Highest relevance is Exact, with declared gains 3/1/0 for Exact/Partial/Irrelevant. The historical ESCI experiment uses 500 US small-version test queries and a fixed union of 10,076 products with 10,134 judgments [5, 15]. We report judged-relevant recall; incomplete judgment coverage prevents claims about every truly relevant product.

The extension freezes the existing WANDS development split: 96 queries selected by SHA256 order and 384 held-out queries. Historical test results have already been inspected, so these are post-hoc fixed-strategy evaluations, not pristine confirmation. Only development labels choose the global rule and the best added configuration among the rules, centroid2/4, and max2/4. The chosen configuration transfers unchanged to MiniLM and ESCI; $m=7$ is a saturation check and cannot replace the winner. All screening results, including losses, remain available.

A further 100 ESCI query IDs absent from the historical query set are frozen before retrieval. Their judged products are added to the historical union to define a new common catalog for all compared methods. This evaluates newly sampled queries, while acknowledging prior access to the source dataset. Absolute recall

differences from the old catalog cannot be attributed to generalization alone because competition changes. No product-disjoint or unseen-category claim is made. The new catalog contains 12,019 products and 1,974 judgments. Recall/cNDCG eligible query counts are 308/383 for held-out WANDS, 499/500 for historical ESCI, and 100/100 for the new queries.

5.2 Models and Reconciliation

We retain pinned MiniLM, BGE, GTE, and the historical product-specific encoder for sensitivity evidence; exact model identifiers and revisions are in the artifact. New strategy selection uses frozen BGE; MiniLM tests transfer to another encoder family without retuning. BGE uses CLS and 512 tokens, MiniLM mean pooling and 256 tokens. Query prefixes remain empty, matching the historical condition; BGE v1.5 supports instruction-free use although its model card recommends an instruction for retrieval. CUDA weights are FP16 with FP32 pooling, normalized vectors, and exact stable catalog-order ranking. Models are not retrained.

The audit found that historical ESCI gains interchanged Substitute and Complement. The benchmark paper specifies gains 1, .1, .01, and 0 for E, S, C, and I [15]. Its training code agrees, but the pinned repository’s qrels/evaluation-shell combination is internally inconsistent with that definition. We therefore recompute effectiveness from unchanged ranks under the paper’s gains and retain the historical values in a reconciliation artifact. Highest-label recall, ranks, and VI do not depend on that swap. The earlier mitigation join repair is preserved: cNDCG retains every positive-IDCG query, independently of highest-label availability.

5.3 Uncertainty, Cost, and Reranking

Primary strategy comparisons use query-macro Recall@100 and paired query bootstrap with 10,000 resamples. We report fixed contrasts against Original and hybrid, the selected rule against Original, and direct set-mean–canonical comparisons. Confidence intervals spanning zero do not establish equivalence; no non-inferiority margin is claimed. Diagnostic rank strata use common-query support and clustered query uncertainty, rather than treating all pairs as independent.

We reuse full rankings for $K = 20, 50, 100, 200, 500, 1000$. Local measurements use an Intel i7-13700H CPU and RTX 4060 Laptop GPU with 8 GB memory. Cost separates dense-vector/sparse-posting storage, query encoding, and exact scoring/sorting on four CPU threads. Repeated timing uses the first 50 queries, a warm-up, and five measured repetitions. Encoding costs and distinct-view counts are retained in the artifact. These are local measurements, not ANN or commercial-scale latency estimates.

The new reranker experiment uses the pinned MiniLM cross-encoder with one fixed M2 canonical text per product, joint 256-token input, $K = 50, 100, 200$, and a final Top-20. Candidate-union caching avoids repeated inference. Returned-list condensed NDCG removes unjudged items from those final results but normalizes against all judged ideal gains; it is distinguished from full-catalog cNDCG. The historical experiment, which also varied reranker input, is retained as a separate condition. Its highest-relevance Top-100 memberships survive reconciliation even where floating-point and tie handling alter other candidate positions.

5.4 Optimization-Robustness Extension

After a development-only smoke test, we freeze eight highest-label-eligible queries from WANDS held-out and eight from historical ESCI by dataset-specific SHA-256 order. The sample contains 310 WANDS and 111 ESCI query–target pairs; every highest-label target for each selected query is retained. Sampling does not use crossing status, rank, length, or generated output. This bounded target-only design avoids generating either complete catalog and limits inference while preserving all selected targets.

Both prompts use Qwen2.5-0.5B-Instruct at a pinned revision [14], FP16, Transformers 4.55.4, and a 512-token output budget. Greedy primary generation records seed 20260907; stochastic controls use temperature .7, top- p .9, and seeds 20260981–20260983. The cache key includes the actual input, complete prompts, model revision, decoding settings, and replicate ID. Generated text is reused unchanged for BGE and MiniLM encoder transfer. This small local rewriter makes factual failures and limited generality plausible; it was selected for an executable no-fee audit, not as a state-of-the-art GEO model.

The primary endpoint is query-macro target inclusion@100; inclusion@20 is secondary. Six contrasts per dataset/encoder are frozen: G_1 –Raw, $C \rightarrow G_1 - G_1$, $C \rightarrow G_1 - C$, and the corresponding three for G_2 . We use 10,000 query-cluster bootstrap draws and paired sign flips, with Holm correction within each cell. Target-only interventions do not define a common full ranking, so we do not report cNDCG for them. Generation time, output length, retriever truncation, fact checks, and offline embedding time are retained with rank-level outputs.

6 Results

6.1 RQ1: How Does Representation Affect Relevant-Product Inclusion?

Fact-equivalent orders repeatedly move judged-relevant products across operational cutoffs. Across three general encoders and two datasets, catalog-wide pair-micro VI@20 ranges from 4.42% to 13.26%; the product-specific encoder yields 6.07% on WANDS and 7.22% on ESCI. Figure 2 reports query-macro estimates and uncertainty. These values describe crossing somewhere in a finite family of seven representations, rather than the expected loss from one arbitrary update.

The stricter target-only intervention fixes every competitor in C0 and replaces one relevant product at a time. WANDS pair-micro VI@20 is 12.72%, 8.84%, and 11.39% for MiniLM, BGE, and GTE; ESCI values are 3.11%, 3.16%, and 6.68%. Figure 1 is drawn from this condition: only the target order changes between ranks 5 and 1,527. This supplies the controlled causal statement, while the catalog-wide experiment represents platform deployment where competitors change together.

The effect remains without encoder truncation. Among 8,536 WANDS Exact pairs whose seven complete inputs fit BGE, query-macro VI@20 is 8.69% (CI [7.15,10.39]). Instability also omits stable failure: at Top-100, WANDS BGE has 8,384 highest-label pairs always included, 3,856 sometimes included, and 13,230 never included. ESCI has 3,995, 44, and 395, respectively. A method can therefore

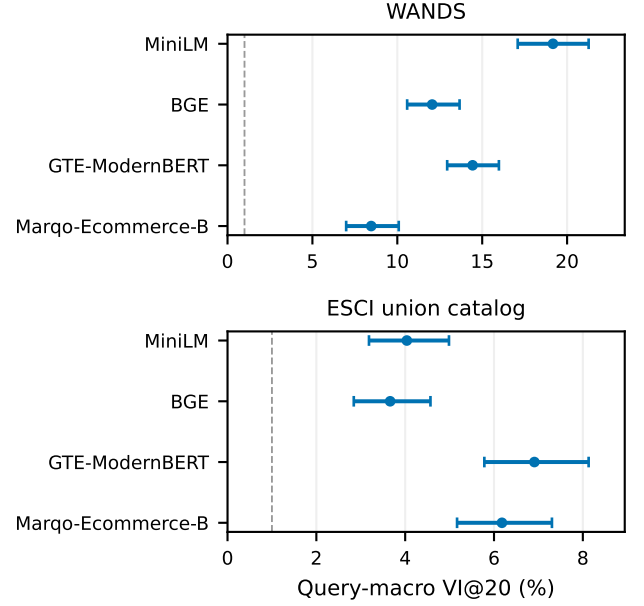


Figure 2: Catalog-wide query-macro VI@20 with query-bootstrap 95% CIs. The dashed 1% line is a historical study threshold.

achieve zero VI by retrieving every relevant product or by omitting the same products consistently.

6.2 RQ2: Do Visibility Rewriters Withstand Attribute Order?

We next apply both frozen adapted E-GEO rewriters after each raw order. Figure 3 separates mean target inclusion from order sensitivity on the fixed eight-query samples. With BGE on WANDS, Raw has 32.1% mean inclusion@100 and 14.8% VI. The heuristic has similar mean inclusion (32.3%; paired change +0.25 points, CI [−9.82,14.42]) but more than doubles VI to 31.6%. The technical prompt reaches 31.6% mean inclusion and 23.2% VI (change −0.47 points, [−3.25,2.19]). Thus neither rewriter improves the primary WANDS outcome, and both are more sensitive to order.

On ESCI, Raw has 87.5% mean inclusion and 3.0% VI. The heuristic falls to 74.7% and raises VI to 17.2% (change −12.74 points, [−23.49,−4.45]); the technical prompt falls to 82.6% with 10.0% VI (−4.89 points, [−9.43,−1.45]). These are operational pipeline outcomes, not pure order effects, because most rewrites fail the fact check. Still, the deployment question is answered directly: an unguarded query-blind rewriter can amplify, rather than absorb, incoming representation variation.

Transition counts expose churn hidden by an average. Across 2,170 WANDS order–target trials at Top-100, the heuristic rescues 346 Raw misses but newly removes 176 Raw hits; on ESCI it rescues only 5 and newly removes 95 across 777 trials. The technical prompt rescues/removes 116/134 on WANDS and 1/36 on ESCI. Canonical-first heuristic outputs still replace 331/166 WANDS decisions and 0/106 ESCI decisions relative to Raw, even though their VI is zero.

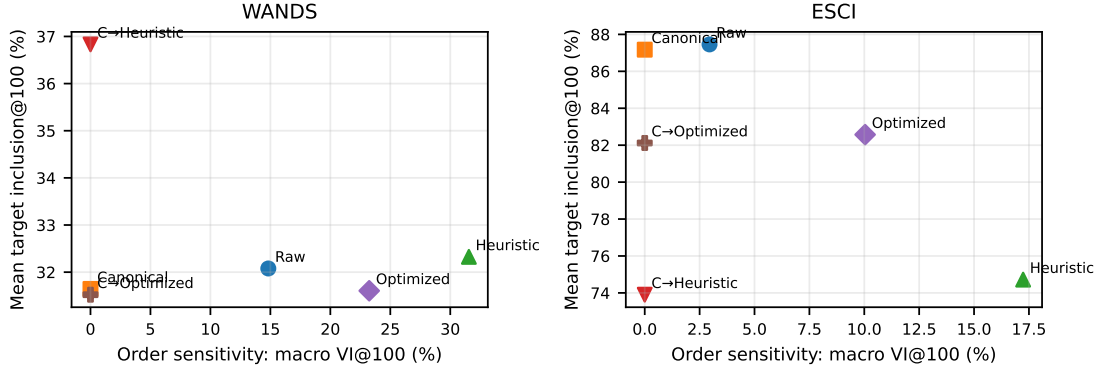


Figure 3: Primary target-only BGE results at $K = 100$. Each point reports query-macro mean inclusion and VI over seven input orders. Canonical-first inputs coincide across orders by construction and deterministic caching; zero VI therefore does not show that the generator itself is stable. Detailed values are in Appendix Table 1.

Table 1: Primary target-only BGE results at $K = 100$. Mean inclusion, VI, and robust coverage (included under every tested order) are query-macro percentages; Never is pair-micro. This is a family of single-target counterfactual indexes, not one catalog Recall.

Method	WANDS				ESCI			
	Mean	VI	Robust	Never	Mean	VI	Robust	Never
Raw	32.1	14.8	23.3	68.1	87.5	3.0	86.4	11.7
Canonical	31.6	0.0	31.6	75.2	87.2	0.0	87.2	13.5
Heuristic	32.3	31.6	19.0	48.7	74.7	17.2	66.0	18.0
C→Heuristic	36.8	0.0	36.8	70.0	73.9	0.0	73.9	27.0
Optimized	31.6	23.2	20.2	69.4	82.6	10.0	77.1	13.5
C→Optimized	31.5	0.0	31.5	78.4	82.1	0.0	82.1	18.0

Counts are pair-micro events, whereas reported mean changes weight queries equally; their signs therefore need not coincide. The practical point is unchanged: net inclusion alone conceals which relevant products gain and lose access.

Generation noise is a material alternative explanation. In the frozen eight-product stochastic control, WANDS same-input Top-100 crossing ranges from 16.7% to 50.0% across method/input cells, comparable to the within-replicate order VI of 16.7%–50.0%. ESCI has only two products per cell and is correspondingly uninformative. We therefore do not attribute the stochastic ranks solely to input order. The main experiment uses greedy decoding, but deterministic decoding is an implementation condition rather than a general stability guarantee.

6.3 RQ3: Are Canonicalization and Factuality Guards Sufficient?

Canonicalizing before generation makes all seven inputs identical. With deterministic content-addressed caching, $C \rightarrow G$ consequently has zero measured VI. It can nevertheless change inclusion. Relative to Canonical, the heuristic point estimate is +5.19 points on WANDS (CI $[-4.05, 16.13]$) and -13.27 on ESCI ($[-24.60, -4.06]$); the technical prompt changes it by -0.12 ($[-3.37, 4.01]$) and -5.05 points ($[-10.61, -0.96]$). Canonicalization controls incoming order;

it does not ensure that the subsequent rewrite preserves facts or improves retrieval.

Table 3 explains why the favorable WANDS heuristic point estimate is not credible evidence of fact-preserving optimization. No WANDS output passes the conservative field-level checker. On ESCI, only 7.6% of raw-order and 8.1% of canonical-input heuristic outputs pass; no technical-prompt output passes. The technical prompt reaches the 512-token output cap for 65.8%–80.3% of cells, while the heuristic truncates 9.9%–17.9%. WANDS source records also contain conflicting fields that a valid rewrite must retain rather than silently reconcile. The checker is deliberately strict and is not human ground truth, but the common-valid support is only two ESCI products for the heuristic and empty elsewhere, far too small for an effectiveness claim.

The component audit shows that failure is not only truncation. For WANDS raw-order heuristic outputs, 99.7% omit at least one checked attribute value, 98.8% omit a source number, 70.7% omit a source color, and 21.1% introduce a new number; 83.2% of evaluated rows originate from products containing a source-field conflict. ESCI has no such conflict flag, yet its heuristic omits a checked attribute in 84.9% and a number in 75.7%, while adding a number in 18.3%. The technical prompt adds numbers in 62.6% of WANDS and 65.9% of ESCI raw-order outputs. These rules deliberately count

Table 2: Prespecified BGE contrasts in query-macro target inclusion@100 (percentage points; paired query-bootstrap 95% CI). Holm-adjusted randomization results are in the artifact. Intervals crossing zero do not establish equivalence.

	G_1 -Raw	$C \rightarrow G_1 - G_1$	$C \rightarrow G_1 - C$	G_2 -Raw	$C \rightarrow G_2 - G_2$	$C \rightarrow G_2 - C$
WANDS	+0.2 [-9.8,+14.4]	+4.5 [-1.7,+12.5]	+5.2 [-4.0,+16.1]	-0.5 [-3.2,+2.2]	-0.1 [-2.9,+2.8]	-0.1 [-3.4,+4.0]
ESCI	-12.7 [-23.5,-4.5]	-0.8 [-4.8,+4.3]	-13.3 [-24.6,-4.1]	-4.9 [-9.4,-1.4]	-0.5 [-6.2,+4.4]	-5.1 [-10.6,-1.0]

Table 3: Adapted-generator factuality and local cost. Strict pass uses the frozen rule checker; Exec. and Guard are fallback rates. Minutes are unique cached batched GPU generation time; monetary API cost is zero.

Data	Method	n	Pass%	Exec.%	Guard%	Tok.	min
WANDS	G1	2170	0.0	17.9	100.0	366	171.7
WANDS	CG1	310	0.0	14.5	100.0	400	25.1
WANDS	G2	2170	0.0	72.7	100.0	501	315.6
WANDS	CG2	310	0.0	80.3	100.0	508	44.6
ESCI	G1	777	7.6	12.9	92.4	324	19.3
ESCI	CG1	111	8.1	9.9	91.9	330	1.3
ESCI	G2	777	0.0	67.2	100.0	504	27.7
ESCI	CG2	111	0.0	65.8	100.0	505	1.8

any missing constrained value, so they should be read as error flags rather than calibrated semantic-quality probabilities.

The prespecified guard returns raw canonical text for truncation, empty output, or any fact-check failure. Consequently, every guarded WANDS and technical-prompt cell equals Canonical. For ESCI heuristic+BGE, guarded mean inclusion@100 is 86.2% versus 87.2% for Canonical and VI is zero; with MiniLM it is 75.3% versus 75.6% and VI is 1.0%. The residual MiniLM sensitivity comes from the few order-specific outputs that pass. The guard prevents unsupported text from reaching retrieval, but supplies no improvement over canonical preprocessing.

Full-catalog canonical results provide the deployable comparison that target interventions cannot. Recall@100 changes by +0.07 points on WANDS+BGE, +0.07 on ESCI+BGE, and +0.28 on ESCI+MiniLM, with all three intervals spanning zero. WANDS+MiniLM falls by 1.83 points (CI [-3.17,-0.64]). Canonical sorting is therefore a simple structural defense, but its relevance cost is encoder-dependent and intervals containing zero do not prove equivalence.

6.4 RQ4: What Transfers, and What Does Robustness Cost?

The generated text transfers without regeneration to MiniLM, but the direction does not stabilize. On WANDS, the heuristic changes inclusion@100 by -3.77 points relative to Raw (CI [-14.36,6.28]); the technical prompt changes it by +1.62 ([-2.68,6.37]). On ESCI, corresponding changes are +0.72 ([-3.78,5.72]) and -0.89 points ([-4.30,2.49]). No cell supports a consistent cross-encoder gain. The earlier development-selected appearance-first rule similarly loses 1.75 points on WANDS MiniLM ([-3.00,-0.63]), reinforcing that representation choices do not transfer automatically.

The local generator incurred no API charge, but the frozen primary outputs required about 10.1 GPU-hours of measured generation, plus 1.4 hours for the stochastic control. The technical prompt accounts for most compute because its outputs approach the cap.

This exceeded the planned eight-hour generation stop; the deviation is disclosed because completion was not selected by observed ranks and no sample, prompt, or outcome was changed. In contrast, raw canonical sorting needs no language-model inference.

Platform-side alternatives retain different tradeoffs. Equal-weight hybrid retrieval improves Recall@100 from 0.632 to 0.681 on WANDS and from 0.910 to 0.915 on ESCI. A four-view centroid raises WANDS recall by 1.22 points (CI [0.34,2.29]) but is not monotonically improved by seven views. The invariant set-mean representation has small, statistically inconclusive historical cNDCG changes, yet lowers new-query ESCI Recall@100 by 2.03 points ([-3.15,-0.90]). Finally, fixed-text reranking changes WANDS final Recall@20 by +0.66 points at a 100-candidate budget ([-0.03,1.58]); expanding Original from 100 to 500 candidates yields +1.13 ([0.08,2.41]). Reranking can reorder candidates, but it cannot recover products it never receives.

7 Discussion and Limitations

7.1 What the Optimization Audit Establishes

The experiment separates a content intervention from its ranking outcome. Applying a query-blind rewrite to seven fact-equivalent inputs does not make the resulting retrieval robust: the two adapted baselines increase VI, and their effectiveness varies by dataset and encoder. Canonicalization before rewriting blocks incoming-order variation under deterministic caching, but the subsequent rewrite can still reduce inclusion. Once a strict factuality guard is imposed, nearly every output falls back to canonical text. The defensible engineering result is consequently modest: normalize raw atoms before indexing, verify generated text against the complete fact inventory, and evaluate recall as well as stability.

This finding does not overturn E-GEO. Its published experiment rewrites one item within a fixed ten-candidate reranking set with GPT-4.1; ours adapts two released prompts to a 0.5B local model and asks whether products enter a dense first-stage pool. The poor fact retention and frequent 512-token truncation primarily characterize this adaptation. They nevertheless expose a deployment requirement that prompt-level instructions alone do not satisfy: a product rewriting pipeline needs explicit factuality and failure handling before rank changes can be interpreted as relevance-preserving visibility gains.

The generation control also changes what can be claimed causally. Raw permutations are verified by atom equality, so their target-only rank changes isolate representation. Free-text generations are not equivalent by construction. Under stochastic decoding, the same input sometimes crosses Top-100 across replicates as often as different input orders do. These sources must be modeled separately; pooling all outputs would over-attribute generator noise and semantic drift to serialization.

7.2 From Retrieval Visibility to User Exposure

Visibility here means eligibility at an offline retrieval cutoff. It is not observed exposure, clicks, satisfaction, purchase, or welfare, and relevance labels are not objective product quality. The shopping-agent motivation explains why candidate inclusion matters but does not substitute for an agent or user evaluation. The mechanism may arise wherever unordered records are serialized for retrieval, including jobs, hotels, restaurants, and events; neither its existence nor magnitude in those domains is established here.

7.3 Limitations and Evaluation History

The evidence covers two English public datasets, synthetic fact-preserving permutations, fixed catalogs, and exact offline retrieval. ESCI judgments are incomplete and its historical common catalog differs from official small candidate pools. The optimization audit contains only eight frozen queries per dataset; query-cluster intervals are accordingly wide, and the two-product common-valid subset supports no effectiveness inference. The source datasets and earlier test results had already been inspected before this post-hoc extension, although sample selection and methods were frozen before extension ranks were read.

The adapted generator differs from the original model, data, objective, and candidate setting. The automated fact checker is conservative, incomplete, and not human ground truth; source conflicts make WANDS especially demanding. Greedy decoding and deterministic caching are implementation choices. The stochastic control is small. The measured generation exceeded the planned eight-hour stop by about 3.5 hours across primary and noise runs; no method, sample, or stopping decision used ranking outcomes, but this remains a protocol deviation. We test no paid API, trained CHARM model, ANN index, commercial system, live shopping agent, or user behavior. Confidence intervals containing zero do not establish equivalence, and no fairness or disparate-impact conclusion follows from these results.

8 Conclusion

Fact-equivalent attribute orders can determine whether a relevant product enters a neural retrieval pool, even when only that product changes. Two adapted visibility rewriters do not repair this problem: they amplify order sensitivity, often alter or omit source facts, and do not transfer consistently across datasets or encoders. Canonical preprocessing removes incoming-order variation, while a factuality guard makes the tested generation pipelines behave almost entirely like canonical text. Because canonicalization itself can reduce recall, robustness must be assessed jointly with relevant-item inclusion. The practical lesson is to treat product attributes as structured facts, verify every generated representation, and evaluate candidate loss before downstream ranking or agents are credited with improved visibility.

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A Optimization-Robustness Details

The frozen protocol and exact query/product IDs precede the extension’s ranking stage, but do not preregister the historical study. Every primary table below uses the same 310 WANDS and 111 ESCI target pairs. Raw and generated rows are families of independent target-only counterfactual indexes; they are not catalog Recall. Canonical-first rows repeat by construction because their actual model input and deterministic cache key are identical.

Operational and valid-output populations. The main operational analysis replaces generation errors and length-truncated text with canonical input, while allowing nonempty fact-check failures to expose the raw adapted pipeline. The guarded variant additionally replaces every fact-check failure. Table 7 requires all seven raw-order outputs and the canonical-input output to pass for the

same product; this leaves two ESCI heuristic products and no other support. Empty cells are retained rather than silently dropped. Fact-component, truncation, fully-fitting, transition, and rank-level files accompany the artifact.

Generation variation. Table 6 computes order VI separately inside each stochastic replicate. Its same-input columns instead hold C0 or canonical text fixed and vary only generation. These quantities are deliberately not pooled. Each WANDS cell contains six query–product pairs and each ESCI cell two; the control diagnoses confounding but does not estimate a stable population rate.

B Preserved Historical Controls

All Phase I–IV tables, figures, ranks, and archived manuscripts remain in the repository. The current main paper retains their validated cross-model VI, fully-fitting, persistent-omission, hybrid, centroid, set-mean, transfer, candidate-budget, and reranking results. Tables 9 and 10 retain the fixed view-count and mitigation results most relevant to the new audit. Full screening, timing, transition, and reconciliation detail remains in the Phase IV artifact.

The motivating query is *turquoise chair* (WANDS query 162); the Exact-relevant item is product 34536. Its source title and attributes contain a color discrepancy, which the audit preserves rather than resolving. The complete target-only ranks below are an inspectable post-hoc example, not an effect-size estimate.

Model cache keys include content and revision hashes. Exact ranking uses stable catalog-order ties and removes the old target before counterfactual insertion. The reproduction package records prompts, all generated text, fact checks, hashes, per-order ranks, query-cluster contrasts, and protocol deviations. No ANN approximation, paid API, or outcome-based output selection is used.

Table 4: Query-macro target inclusion@100 for every tested order. Canonical and canonical-first rows repeat by construction.

Data	Encoder	Method	C0	C1	C2-1	C2-2	C2-3	C2-4	C2-5
ESCI	bge_base	C→Heuristic	73.9	73.9	73.9	73.9	73.9	73.9	73.9
ESCI	bge_base	C→Optimized	82.1	82.1	82.1	82.1	82.1	82.1	82.1
ESCI	bge_base	Canonical	87.2	87.2	87.2	87.2	87.2	87.2	87.2
ESCI	bge_base	Heuristic	78.1	71.1	72.5	70.6	75.5	77.5	77.7
ESCI	bge_base	Optimized	82.3	81.8	82.8	82.9	83.2	82.6	82.5
ESCI	bge_base	Raw	88.0	87.8	87.2	86.4	88.0	86.4	88.6
ESCI	minilm	C→Heuristic	75.9	75.9	75.9	75.9	75.9	75.9	75.9
ESCI	minilm	C→Optimized	75.6	75.6	75.6	75.6	75.6	75.6	75.6
ESCI	minilm	Canonical	75.6	75.6	75.6	75.6	75.6	75.6	75.6
ESCI	minilm	Heuristic	75.4	73.2	75.3	75.9	78.4	74.4	75.7
ESCI	minilm	Optimized	74.5	72.8	75.8	71.4	77.4	72.4	72.7
ESCI	minilm	Raw	74.8	75.1	74.3	74.3	74.3	75.6	74.8
WANDS	bge_base	C→Heuristic	36.8	36.8	36.8	36.8	36.8	36.8	36.8
WANDS	bge_base	C→Optimized	31.5	31.5	31.5	31.5	31.5	31.5	31.5
WANDS	bge_base	Canonical	31.6	31.6	31.6	31.6	31.6	31.6	31.6
WANDS	bge_base	Heuristic	36.1	35.1	31.1	27.8	34.8	26.5	34.8
WANDS	bge_base	Optimized	29.8	34.0	29.9	29.3	32.9	30.9	34.4
WANDS	bge_base	Raw	34.9	30.3	29.2	33.6	29.9	33.1	33.5
WANDS	minilm	C→Heuristic	28.0	28.0	28.0	28.0	28.0	28.0	28.0
WANDS	minilm	C→Optimized	24.2	24.2	24.2	24.2	24.2	24.2	24.2
WANDS	minilm	Canonical	23.7	23.7	23.7	23.7	23.7	23.7	23.7
WANDS	minilm	Heuristic	21.3	25.4	19.7	22.2	20.8	6.9	13.6
WANDS	minilm	Optimized	24.6	27.4	22.5	24.3	24.0	23.9	20.9
WANDS	minilm	Raw	28.1	18.6	23.8	21.7	25.8	17.8	20.6

Table 5: Unchanged generated texts transferred to MiniLM. Values are query-macro target inclusion, VI, and robust coverage at 100 (%).

Data	Method	Mean	VI	Robust
WANDS	Raw	22.3	19.1	13.7
WANDS	Canonical	23.7	0.0	23.7
WANDS	Heuristic	18.6	34.5	1.0
WANDS	C→Heuristic	28.0	0.0	28.0
WANDS	Optimized	24.0	15.5	17.9
WANDS	C→Optimized	24.2	0.0	24.2
ESCI	Raw	74.8	1.3	74.3
ESCI	Canonical	75.6	0.0	75.6
ESCI	Heuristic	75.5	14.8	67.3
ESCI	C→Heuristic	75.9	0.0	75.9
ESCI	Optimized	73.9	16.2	66.9
ESCI	C→Optimized	75.6	0.0	75.6

Table 6: Stochastic generation control. Order VI is computed separately within each of three replicates; same-input columns vary only generation. Percentages are pair-micro on the frozen eight-product control.

Data	Method	Order VI reps.	C0 repl. VI	Canonical repl. VI
ESCI	G1	50.0/0.0/50.0	50.0	0.0
ESCI	G2	0.0/0.0/50.0	0.0	0.0
WANDS	G1	33.3/33.3/50.0	33.3	50.0
WANDS	G2	33.3/16.7/33.3	33.3	16.7

Table 7: Common fact-valid support for all seven generated orders. Inclusion values are BGE query-macro at 100 on that common support; empty support is shown explicitly. Automated validity is conservative and is not human ground truth.

Data	Opt.	Products	Rate	Raw	G	C→G
WANDS	G1	0	0.0	–	–	–
WANDS	G2	0	0.0	–	–	–
ESCI	G1	2	1.8	100.0	50.0	50.0
ESCI	G2	0	0.0	–	–	–

Table 8: Pure raw-atom canonical sorting versus Original in full-catalog query-macro metrics (percentage points; paired query-bootstrap 95% CI). This is deployable catalog Recall, unlike target-intervention inclusion.

Data	Encoder	Metric	Δ [95% CI]
WANDS	bge_base	Recall@20	-0.0 [-0.5,+0.4]
WANDS	bge_base	Recall@100	+0.1 [-1.5,+1.6]
WANDS	bge_base	cNDCG@10	-0.2 [-0.6,+0.3]
WANDS	minilm	Recall@20	-2.1 [-4.0,-0.3]
WANDS	minilm	Recall@100	-1.8 [-3.2,-0.6]
WANDS	minilm	cNDCG@10	-0.2 [-0.8,+0.5]
ESCI	bge_base	Recall@20	-0.3 [-0.7,+0.1]
ESCI	bge_base	Recall@100	+0.1 [-0.0,+0.2]
ESCI	bge_base	cNDCG@10	+0.1 [-0.2,+0.5]
ESCI	minilm	Recall@20	-0.5 [-1.0,+0.0]
ESCI	minilm	Recall@100	+0.3 [-0.0,+0.8]
ESCI	minilm	cNDCG@10	+0.0 [-0.3,+0.3]

Table 9: Fixed BGE view-count checks on held-out WANDS and historical ESCI. Differences are Recall@100 percentage points from Original. These checks do not change the selected strategy.

Data	Method	R100	Δ	95% CI
WANDS	centroid 2	0.640	+0.78	[-0.11,+1.79]
WANDS	max 2	0.640	+0.83	[+0.02,+1.80]
WANDS	centroid 4	0.644	+1.22	[+0.34,+2.29]
WANDS	max 4	0.639	+0.69	[-0.30,+1.80]
WANDS	centroid 7	0.642	+0.99	[+0.02,+2.07]
WANDS	max 7	0.642	+0.98	[-0.11,+2.23]
ESCI	centroid 2	0.911	+0.09	[-0.08,+0.25]
ESCI	max 2	0.911	+0.11	[-0.07,+0.28]
ESCI	centroid 4	0.910	+0.04	[-0.12,+0.19]
ESCI	max 4	0.911	+0.07	[-0.11,+0.25]
ESCI	centroid 7	0.910	+0.04	[-0.12,+0.18]
ESCI	max 7	0.910	+0.07	[-0.11,+0.25]

Table 10: Reconciled historical mitigation cNDCG@10 and paired differences from Original. ESCI uses E/S/C/I=1/.1/.01/0. WANDS uses 383 positive-IDCG held-out queries; ESCI uses 500.

Data	Method	cN@10	Δ	95% CI
WANDS	Original	0.8247	+0.0000	[+0.0000,+0.0000]
WANDS	Canonical	0.8030	-0.0217	[-0.0333,-0.0105]
WANDS	Factual	0.8178	-0.0069	[-0.0139,-0.0000]
WANDS	Set mean	0.8151	-0.0096	[-0.0194,+0.0003]
WANDS	Late mean	0.8058	-0.0189	[-0.0293,-0.0083]
WANDS	Late max	0.8093	-0.0153	[-0.0266,-0.0045]
WANDS	Late top3	0.8106	-0.0141	[-0.0243,-0.0038]
ESCI	Original	0.7364	+0.0000	[+0.0000,+0.0000]
ESCI	Canonical	0.7423	+0.0058	[-0.0012,+0.0131]
ESCI	Factual	0.7305	-0.0059	[-0.0127,+0.0007]
ESCI	Set mean	0.7326	-0.0038	[-0.0147,+0.0071]
ESCI	Late mean	0.7290	-0.0074	[-0.0181,+0.0031]
ESCI	Late max	0.7297	-0.0067	[-0.0180,+0.0046]
ESCI	Late top3	0.7290	-0.0074	[-0.0181,+0.0031]

Table 11: All orders for the Exact-relevant turquoise chair (MiniLM). Target-only competitors stay in C0. Positive Δ means a worse rank than target C0.

Order	Target-only rank	Δ vs C0	Catalog-wide rank
C0	486	0	486
C1	476	-10	295
C2s1	1,527	+1,041	1,303
C2s2	28	-458	21
C2s3	582	+96	484
C2s4	5	-481	3
C2s5	5	-481	4